

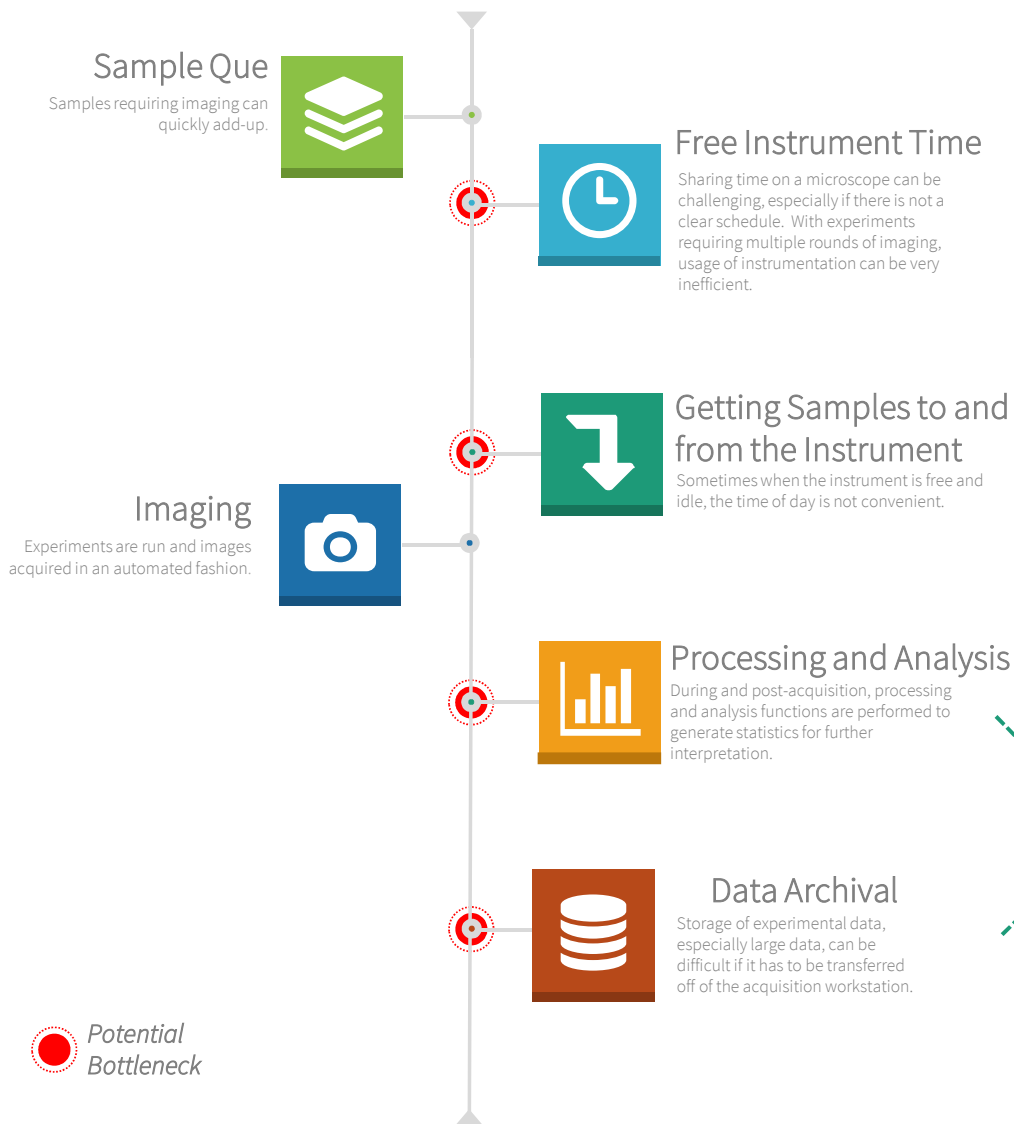


BioPipeline **PLATE**

Microscope Automated Multiplate
Imaging System



Experiment Pipeline

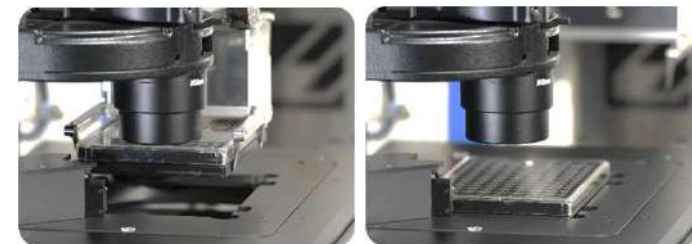


streamlines experiment protocols by introducing organized automation and data storage capabilities.

✓ A scheduler keeps track of plates, owners, and experiment times.



✓ Robotics transfer plates to and from the microscope



✓ A dedicated server can manage offline processing, analysis, and storage.



Key Features



The Perfect Platform for High Content

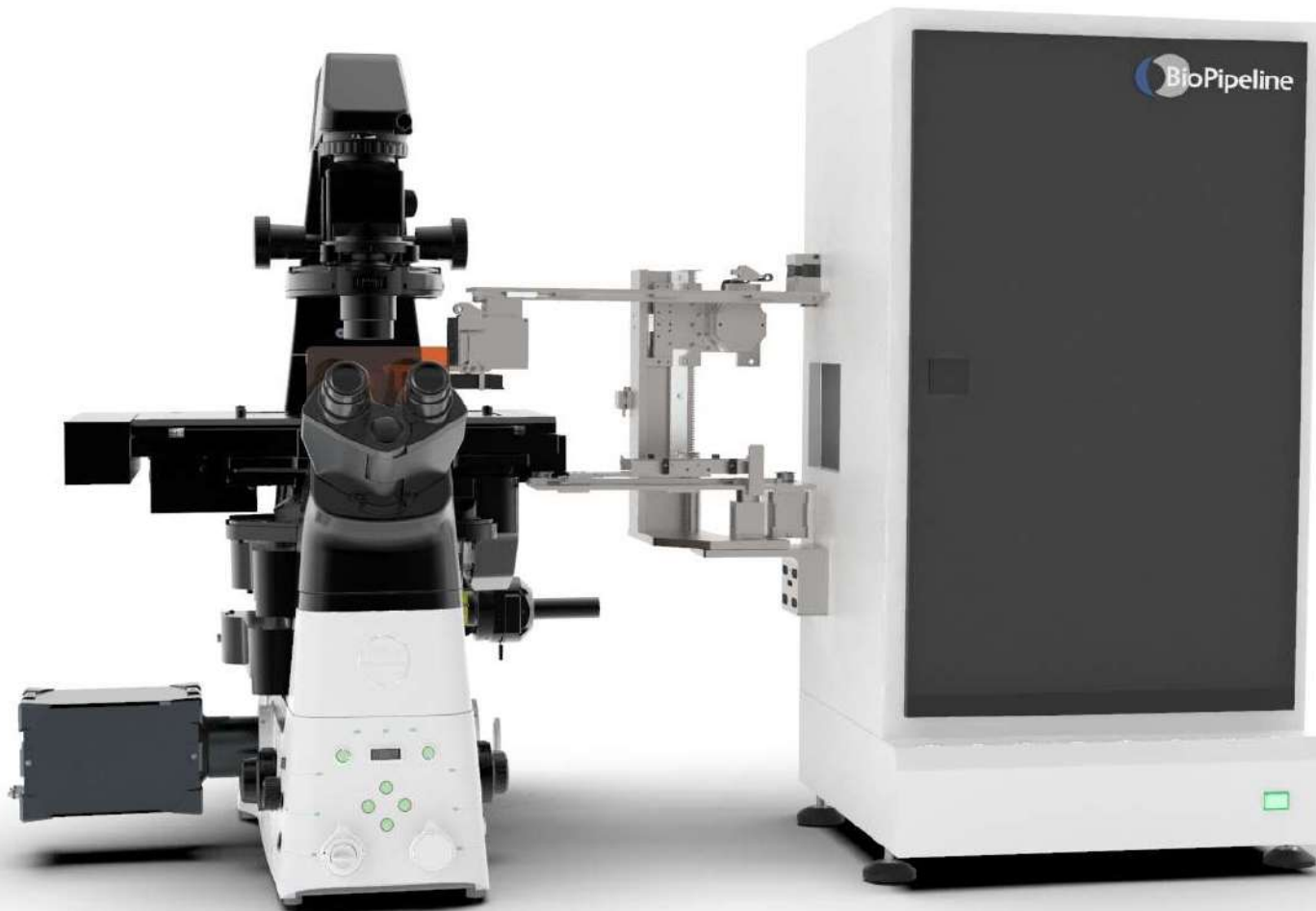
We chose our inverted microscope platform for high content because it was the most flexible: the largest selection of objective lenses, detectors, modalities and applications available to apply to high content acquisition and analysis.

With confocal modalities, 3-dimensional high content at the highest resolution can be acquired.

Also important is the ease of modification or upgrade of the system for new detectors and modalities. With a unified software package across all microscope platforms, data can be easily shared and visualized as well as learning time reduced.



THINK OUTSIDE OF THE BOX SYSTEM



BioPipeline Plate gives users the resolution, flexibility, and power of a research inverted microscope, but the ease-of-use and automation of traditional box-system plate readers.

In addition, it adds **44-plate capacity** and robotic transport from storage to stage top for imaging.

By using the Ti2-E research inverted stand, protocols can include PFS (perfect focus), automated water immersion dispensing, point or field-scanning laser confocal or widefield imaging up to 25mm field of view (FOV), as well as the introduction of illumination devices and schemes for advanced options such as photostimulation.

A microscope platform means easier upgradability when new experiments require new options, and the familiarity of a system with hands-on controls for when you want them.

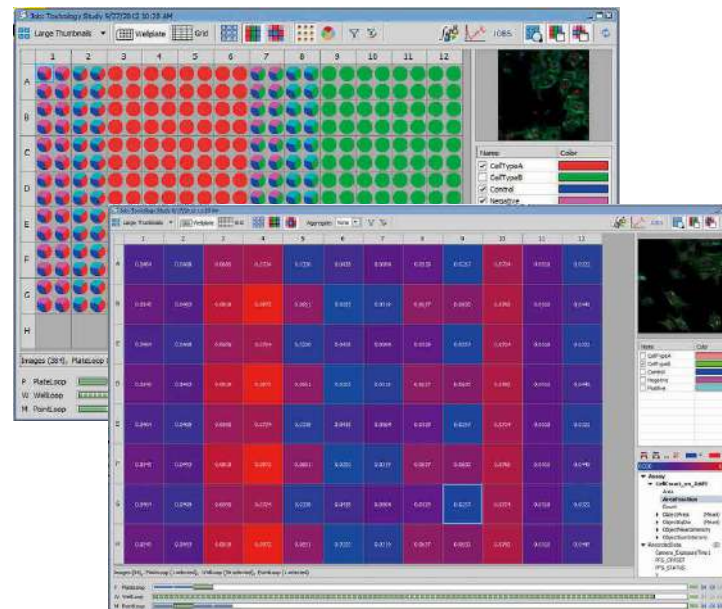
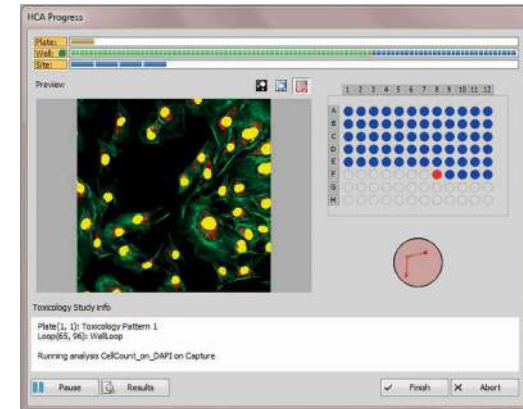
Software



NIS-Elements HC (High Content) allows users to create custom experimental protocols or use turn-key templates to run on multiple plates.

Plate assays and image storage can be shared or specified user by user, and accessed by NIS-Elements Scheduler.

In addition to automated microscopy, NIS-Elements can perform processing and analysis functions including analysis assays either during experiment runs, or later offline on the BioPipeline dedicated server.



NIS-Elements Scheduler keeps track of plates in the incubator and their owners. Owners can schedule time blocks directly within the scheduler to execute JOBS experiment runs.

The NIS-Scheduler then initiates NIS-Elements HC, loads plates, executes experimental runs, returns plates to the incubator, and closes NIS-Elements automatically.

Optionally, the scheduler can email an owner when plates are finished run, or in case of any issues. NIS-Elements can additionally email or SMS text message a user during a plate run with updates defined by the user (e.g., cell counts).

Plate runs can be visualized in a number of methods, including images, labels, and heat maps: giving users an overview of results.

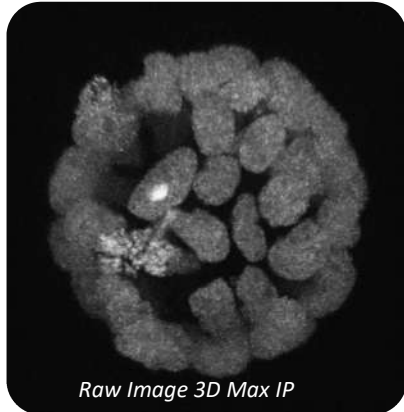
Since analysis can be performed during acquisition, analysis results can be summarized during the experiment or be ready as soon as the experiment completes for viewing.

Image Processing

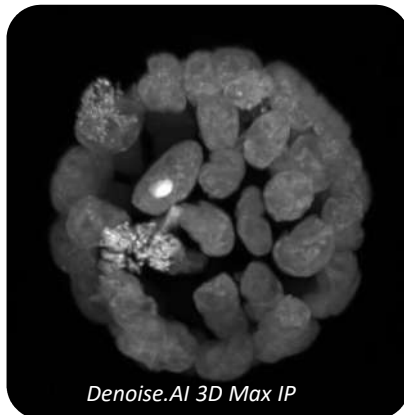
Artificial Intelligence Denoising extends acquisition times

An artificial intelligence neural network trained to recognize and removed Poisson (shot) noise can be applied to confocal images to enhance image details.

Such improvements allow users to collect data at shorter exposures and lower illumination power than ever before, allowing more time points to be collected.

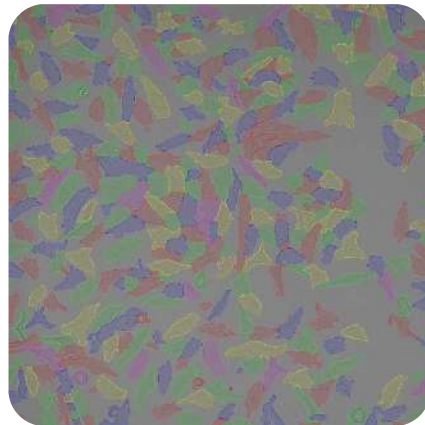
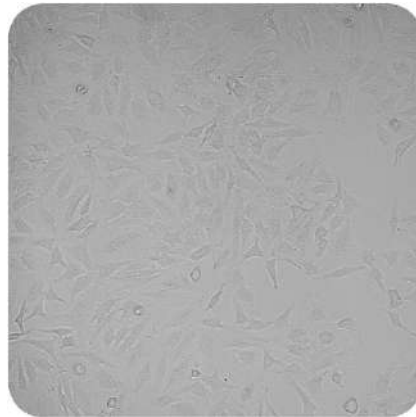


Raw Image 3D Max IP



Denoise.AI 3D Max IP

Label-Free Imaging with Volume Contrast

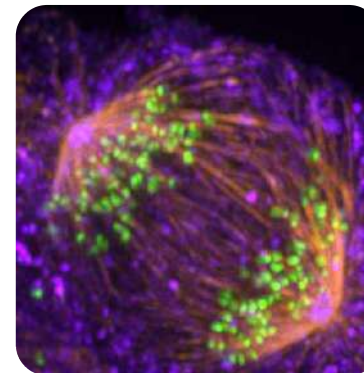
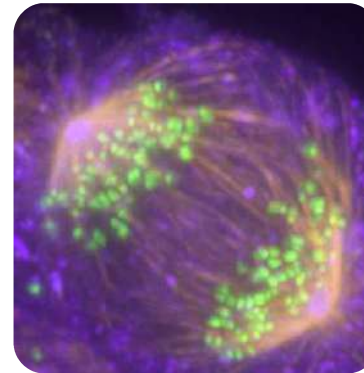


Acquisition of a transmitted light image without contrast (no phase contrast or DIC) can now be used to compute a **Volume Contrast** enhanced result image in which the volumetric shape of objects produces contrast. The contrast image can be used for segmentation.

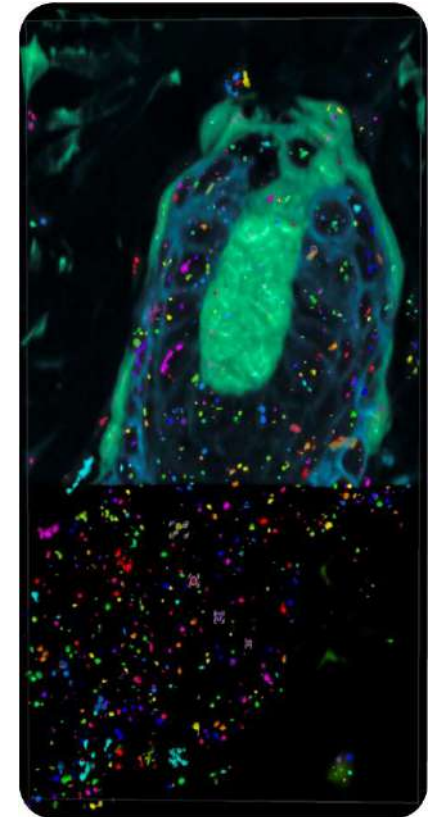
Deconvolution enhances 3D structures

Advantageous to 3-Dimensional high content imaging is the ability to perform **3D Deconvolution** on data sets to improve contrast and sharpness of images. BioPipeline can run deconvolution during acquisition, or automatically on images in target folders on its server post-acquisition.

Deconvolution is supported by GPU-acceleration for maximum speed.



3D Processing and Segmentation



Z stacks can be processed as voxels to incorporate objects in 3 dimensions for preprocessing and enhancement, as well as 3 dimensional thresholding and binary processing. Volumetric measurement information for field and objects can be queried.

Image Analysis



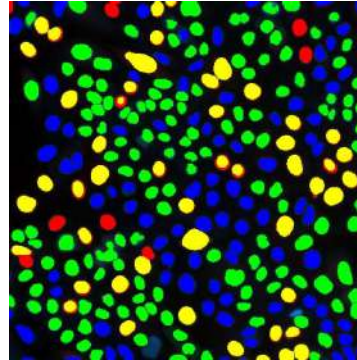
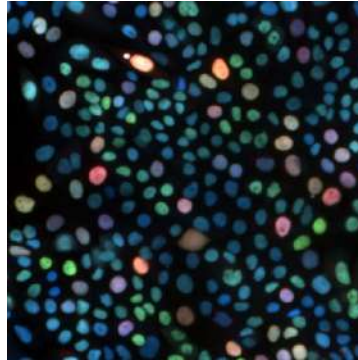
NIS-Elements' **General Analysis** engine employs a powerful processing and analysis toolbox for users.

Dedicated turnkey assays can be applied to image data as well as user-customized assays for specific applications.

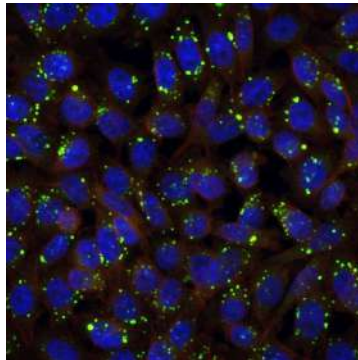
All routines can be executed during acquisition, for example: to modify the course of an experiment, or post-acquisition on the dedicated server.

Assay results can be collated and statistical information displayed, or be exported.

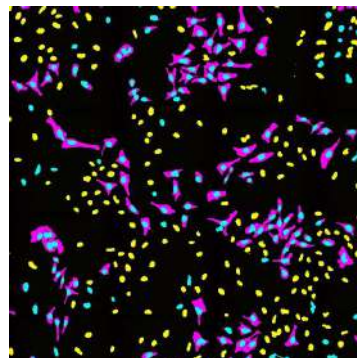
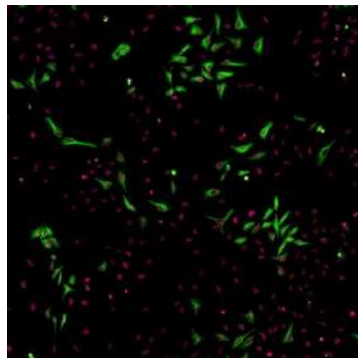
Some example tools are shown here.



The cell labeling tool detects and counts number of cells labeled with colors, or combination of colors. It also provides the intensities of the multiple labels per cell.

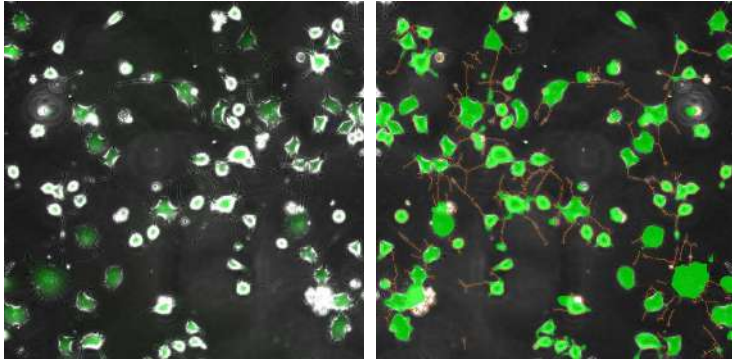


The spot counting tool detects and counts granularity in cells, creating boundaries between cells and reporting number of granules per cell.

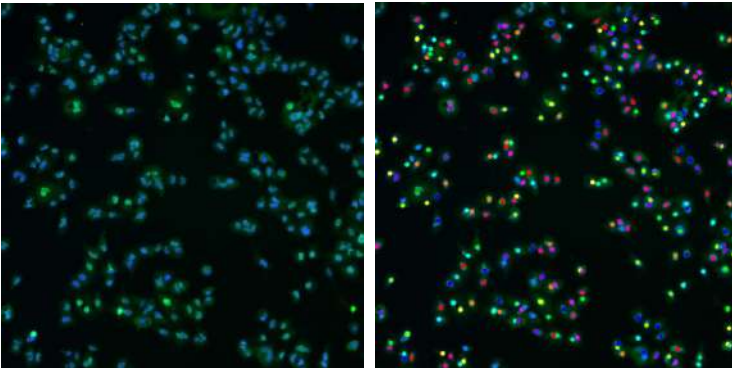


The cell classification tool segments and classifies samples into categories for counting and measurement

Image Analysis

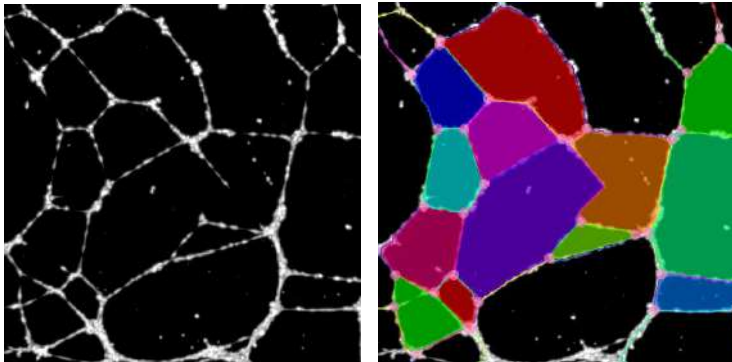


The neurite detection tool highlights neurites and cell bodies, and measures neurite segment length, number of neurites per cell, and branch points per cell.



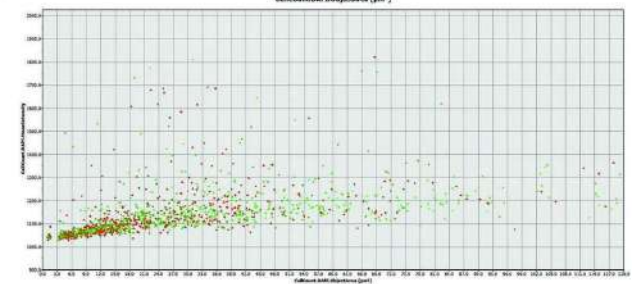
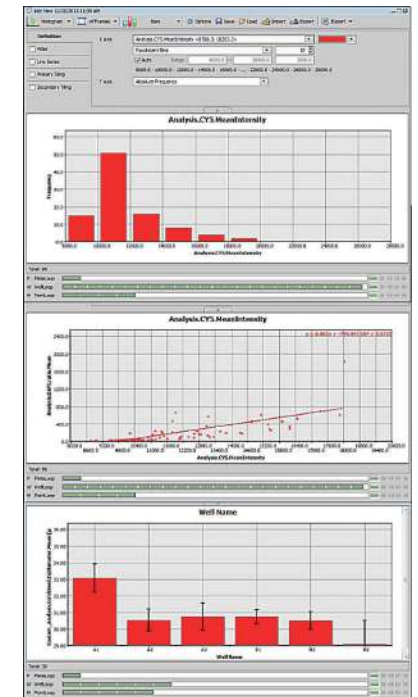
The cell counting tool utilizes low magnification and large FOV to count objects.

This provides accurate and repeatable count information that can be applied across multiple plates.

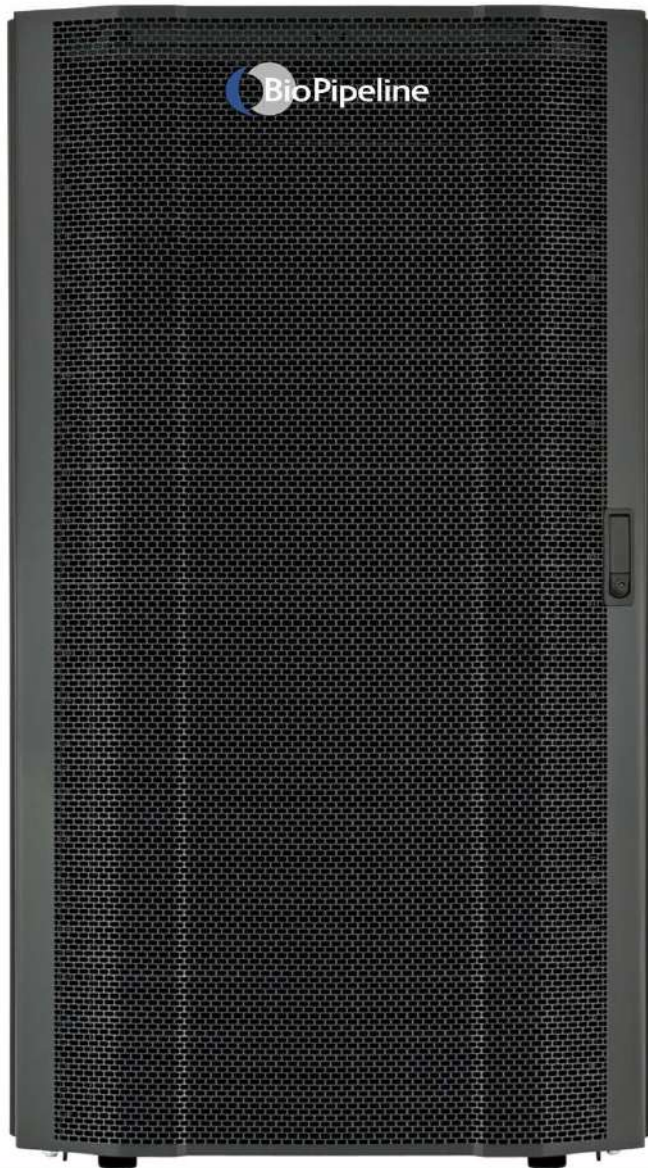


The angiogenesis tool detects tube formation and measures tube number, area, nodes, segment number, and segment lengths.

Analytics can be performed within wells, across wells, or across multiple samples. Statistics and graphs can be generated all within NIS-Elements, or data can be exported in common file formats.



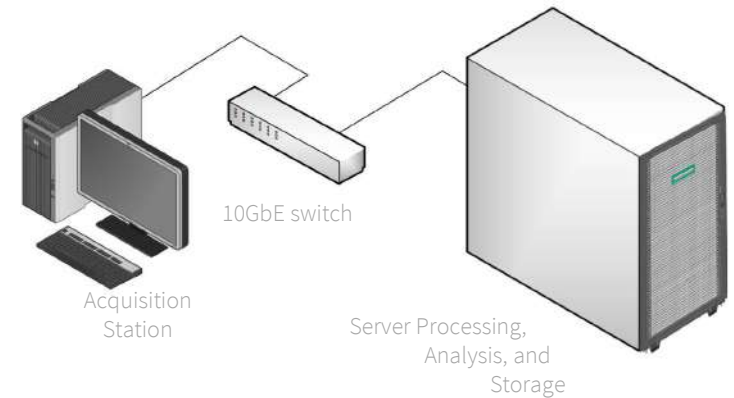
Dedicated Server



Server-side Processing, Analysis, and Storage

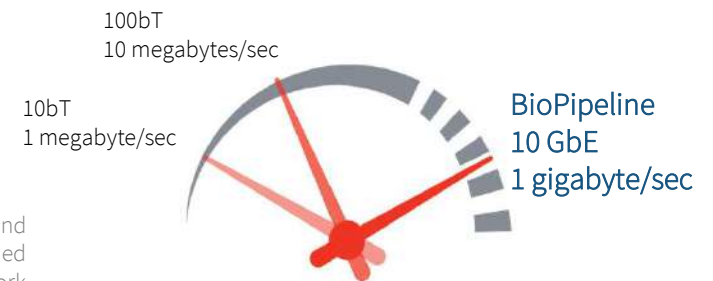
Because data can be saved directly to the server, it means multiple users can view their data while the system is busy acquiring new images. Likewise, the server can perform processing and analysis tasks automatically on designated locations when data is placed there.

Because the server is connected to the imaging workstation by a dedicated switch, slow traffic on the existing network will not affect the transfer speed between acquisition and storage devices.



Data Transfer

BioPipeline's data transfer rates from the imaging workstation to the dedicated server exceed most traditional networks by 10-100x by using dedicated 10 GbE connections.



Storage Capacity

Having **over 40x** the storage capacity of a typical high-end imaging workstation, with expandability for more, the included **BioPipeline Server** with direct dedicated 10GbE network connection allows for more uninterrupted imaging and offline processing and analysis.

Typical Imaging workstation: 6TB

BioPipeline Server: 200TB*

*expandable

Configurations

Full Configuration



Microscope



Robotics



Software and modules



Workstation



DIA and EPI light sources



Server storage



Server-side software



Barcoding



Anti-vibration table

Lite Configuration



Microscope



Robotics



Software and modules



Workstation



DIA and EPI light sources

The Lite configuration of BioPipeline Plate is for users who may not require an antivibration table or bar code preparation, and have other mechanisms in place for server processing or storage.

Select the detector(s) and objective lenses best suited for your experimental pipeline to add to the base packages.

Likewise, if further automated hardware is required, it can be added as well.

And additional software modules can be added as needed.



Detectors



Objectives



Additional software modules



Automatic water immersion



LAPP branches / modules



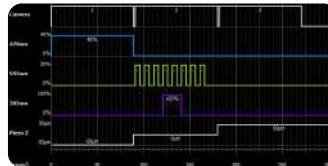
Workstation desk

Options for all configurations



Select from a variety of **objective lenses** specially designed for spherical and chromatic correction, long working distance, or large FOV.

Long-term automated timelapse and multiple plate loading and incubation can be achieved with both air objectives and water immersion objectives.



Select a detector if using the **Open Detector Package**, or add a second detector to the microscope right-side port. NIS-Elements supports all major manufacturers for sCMOS, CCD, and EMCCD detectors and can connect these for hardware triggering of microscope peripheral devices.



Additional **NIS-Elements modules** for specific applications such as co-localization and FRET are also available.

Utilize the **automated water immersion dispenser** for maintaining water immersion for long-term timelapses or imaging large specimens. The dispenser can be used for multiple water immersion objectives on the same nosepiece.



Modify BioPipeline Plate with **additional LAPP modules** to meet experiment requirements, including raster scanning or DMD photostimulation devices and additional epifluorescence illuminators (up to 3 on a 25mm FOV configuration; maximum of 5).

Specifications

Microscope Specifications



Imaging Modalities	Transmitted Light Brightfield Widefield Epifluorescence Point Scanning confocal (option) Field scanning confocal and Optical Pixel Reassignment (options)
Illumination	8 channel Epifluorescence LED LED transmitted Light Laser illumination for confocal sources (option)
Objectives	All air or water-immersion objectives lenses can be used for high content Oil and Silicone immersion objectives can be used for single specimen observation
Imaging Methods	Multidimensional XY, Z, wavelength, timelapse, multi-stage position
Autofocus	Perfect Focus 4 for focus stability; software-controlled autofocus
Hardware Triggering	Supports direct hardware triggering of light sources and shutters
Antivibration Table	3' x 6' x 8"

**see detector brochures for detector specification details*

Plate Loader Specifications

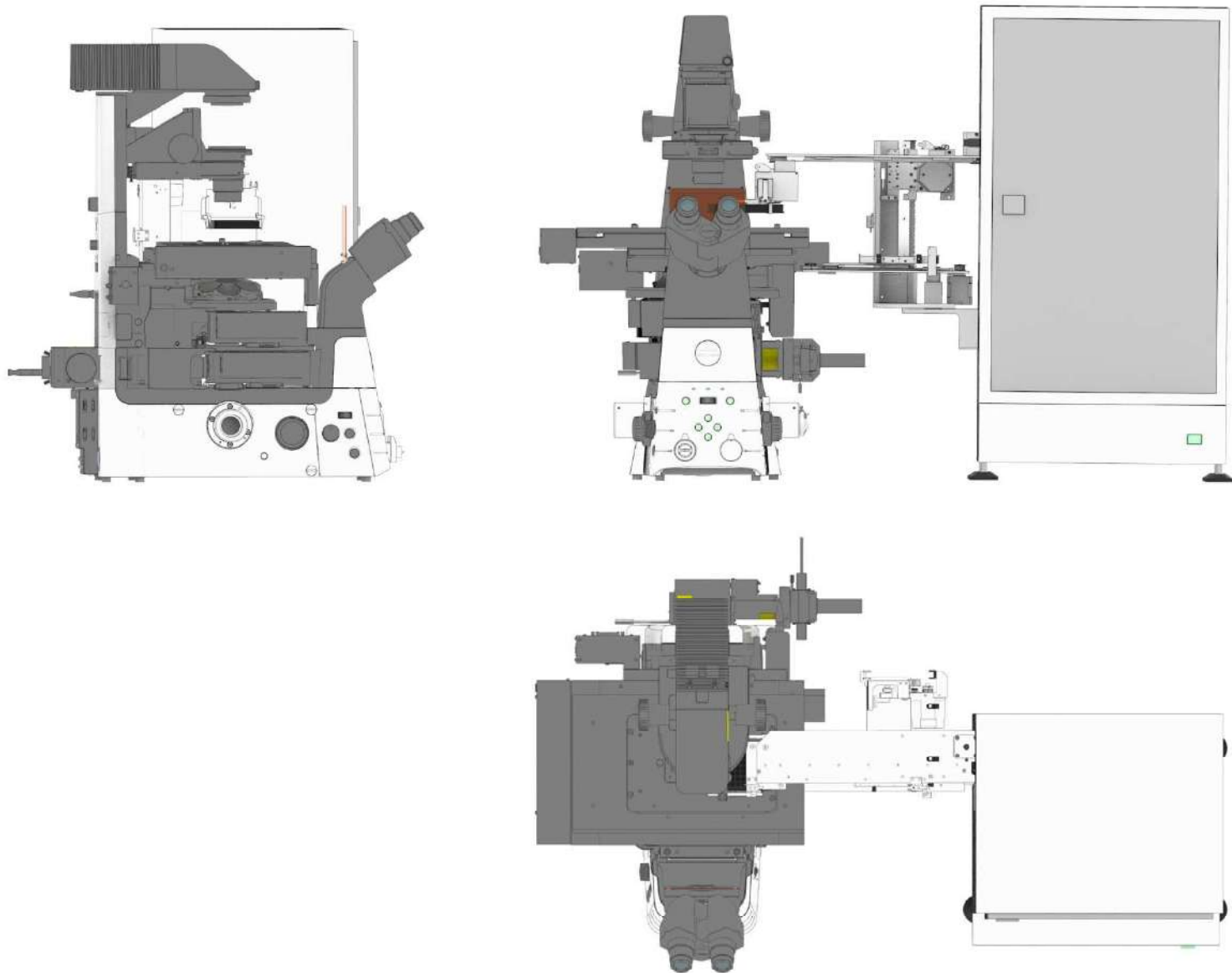


Application	Plate Storage and loading
Capacity	44 plates (128mm x 85mm x 14mm standard) in any format (e.g., 6, 12, 96, 384)
Barcode Format	Code 128
Robotics	Internal shovel robot and external grabber robot
Robotics Speed	3 speed settings for robotic movement acceleration and maximum speed
Communication	RS232 or USB virtual COM port
Power Requirements	100V / 50Hz 115V / 60Hz / 240V / 50Hz ; switchable
Power Consumption	<100W

Server Specifications



Rack Size	22U
Processor	Intel Xeon Silver 412 2.6 GHz
OS	Windows Server
RAM	128 GB
Storage Type	SAS
Storage Capacity	250TB
Network	12x 10GBase-T + 4x 1Gbit/10Gbit SFP+
Support	3 year Foundation Care Service and Support
Power Consumption	600W (average) – 1300W
Options	Tape Backup, 144TB Expansion
Physical Dimensions	(W x D x H) 90.00 x 129.20 x 151.00 cm



Specifications and equipment are subject to change without any notice or obligation on the part of the manufacturer.
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 WARNING	TO ENSURE CORRECT USAGE, READ THE CORRESPONDING MANUALS CAREFULLY BEFORE USING YOUR EQUIPMENT.
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WARNING-LASER RADIATION
AVOID EXPOSURE TO BEAM
CLASS 3B LASER PRODUCT
 Total Power 500mW MAX.
 CW 400~700nm
 IEC/EN60825-1: 2007, 2014

Complies with FDA performance standards for laser products except for deviations pursuant to Laser Notice No. 50, dated June 24, 2007

DANGER-VISIBLE AND INVISIBLE LASER RADIATION AVOID EYE OR SKIN EXPOSURE TO DIRECT OR SCATTERED RADIATION
CLASS 4 LASER PRODUCT
 Total Power 1500mW MAX.
 CW 370~790nm
 IEC/EN60825-1: 2007, 2014

Complies with FDA performance standards for laser products except for deviations pursuant to Laser Notice No.50 dated June 24, 2007.



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